Grade 6 Math – First Mid-Term Revision (Ch 0–3)

Math · Sifana Siddiq · Knowledge City Schools · Answer key ✅

I. Choose the correct option

💡 Review the vocabulary of Ch. 0–3: factors, angles, coordinate plane, volume, decimals, fractions and ratios.

1. In a multiplication problem, what are the numbers being multiplied called?

A Sums
B Differences
C Factors ✓
D Quotients

Why: The numbers multiplied together are factors; the result is the product.

2. If you multiply any whole number by zero, what is the product always?

A One
B The same number
C Zero ✓
D Ten

Why: Anything × 0 = 0.

3. What do you call an angle that measures exactly 90 degrees?

A Acute angle
B Right angle ✓
C Obtuse angle
D Straight angle

Why: A right angle is exactly 90°.

4. An angle that measures less than 90 degrees is called what?

A Acute angle ✓
B Right angle
C Obtuse angle
D Straight angle

Why: Acute = less than 90°; obtuse = more than 90°; straight = 180°.

5. On a coordinate plane, what is the horizontal number line called?

A Y-axis
B X-axis ✓
C Origin
D Quadrant

Why: The x-axis goes left–right (horizontal); the y-axis goes up–down.

6. Volume is the measure of the amount of space occupied by what kind of figure?

A Two-dimensional flat shape
B Three-dimensional solid figure ✓
C A straight line
D A single point

Why: Volume is for 3-D solids (length × width × height).

7. Which of the following is a standard unit used to measure volume?

A Square centimeters
B Cubic units ✓
C Linear inches
D Degrees

Why: Volume is measured in cubic units (cm³, in³, m³).

8. If you multiply a decimal number by a power of ten like 100, which way does the decimal point move?

A To the left
B To the right ✓
C It does not move
D Upwards

Why: × 100 moves the decimal point 2 places to the right (the number gets bigger).

9. In a fraction, what does the denominator at the bottom represent?

A The number of parts being taken or shaded
B The total number of equal parts in the whole ✓
C The remainder of a division problem
D The decimal equivalent

Why: Denominator = how many equal parts the whole is divided into.

10. What do you call a fraction where the numerator is smaller than the denominator?

A Improper fraction
B Mixed number
C Proper fraction ✓
D Whole number

Why: Proper fraction: top < bottom (e.g. 3/5).

11. A theater has 126 seats. So far 100 tickets have been sold. What is the ratio of unsold tickets to total seats?

A 126:26
B 26:126 ✓
C 26 to 100
D 100 to 126

Why: Unsold = 126 − 100 = 26 → unsold : total = 26 : 126.

12. At a library, 27 books, 17 videos, and 8 magazines are checked out in one day. What is the ratio of books to total items?

A 27/52 ✓
B 25:27
C 8:27
D 26 to 52

Why: Total = 27 + 17 + 8 = 52 → books : total = 27/52.

13. Which ratio is equivalent to 36:52?

A 3:12
B 72 to 156
C 9:13 ✓
D 3/4

Why: Divide both by 4: 36 ÷ 4 = 9, 52 ÷ 4 = 13 → 9:13.

14. Salma has 4 onions, 5 lemons, and 6 potatoes. What is the ratio of potatoes to onions?

A 4:5
B four to six
C 6/4 ✓
D six to five

Why: Potatoes : onions = 6 : 4 = 6/4 (order matters).

15. **A bus has 28 seats. Ten seats are empty. What is the ratio of empty seats to the total number of seats?**

A 28:10
B 18:10
C 10 to 28 ✓
D 10/18

Why: Empty : total = 10 : 28.

II-A. Enter each ratio

💡 Part-to-part compares two groups; part-to-whole compares one group to the total. Keep the order given in the question.

16. **3 weasels, 7 toads. Part to Part (weasels to toads):**

A 3:7 ✓
B 7:3
C 3:10
D 10:7

17. **3 weasels, 7 toads. Part to Whole (weasels to animals):**

A 3:7
B 3:10 ✓
C 10:3
D 7:10

Why: Whole = 3 + 7 = 10 animals.

18. **3 weasels, 7 toads. Whole to Part (animals to toads):**

A 7:10
B 3:10
C 10:7 ✓
D 10:3

19. **12 toads, 13 weasels. Part to Part (weasels to toads):**

A 12:13
B 13:12 ✓
C 13:25
D 25:12

20. **12 toads, 13 weasels. Part to Whole (weasels to animals):**

A 13:12
B 12:25
C 13:25 ✓
D 25:13

Why: Whole = 12 + 13 = 25.

21. **12 toads, 13 weasels. Whole to Part (animals to toads):**

A 25:12 ✓
B 12:25
C 25:13
D 13:25

22. **25 roses: 4 are white, 8 are red, the rest are pink. Red to Pink:**

A 8:4
B 8:13 ✓
C 13:8
D 8:25

Why: Pink = 25 − (4 + 8) = 13 → red : pink = 8 : 13.

23. 25 roses: 4 white, 8 red, the rest pink. Pink to Whole:

A 13:25 ✓ B 25:13
C 12:25 D 13:12

24. 25 roses: 4 white, 8 red, the rest pink. Whole to Pink:

A 13:25 B 25:12
C 25:13 ✓ D 12:25

25. 100 roses: 44 are red, 30 are pink, the rest are white. Red, Pink, and White to Whole:

A 74:100 **B 100:100 ✓**
C 26:100 D 74:26

Why: All three colours together = the whole → 100 : 100.

26. 100 roses: 44 red, 30 pink, the rest white. White to Whole:

A 26:100 ✓ B 100:26
C 74:100 D 30:100

Why: White = 100 − (44 + 30) = 26.

27. 100 roses: 44 red, 30 pink, the rest white. Whole to White:

A 26:100 B 100:74
C 100:26 ✓ D 100:44

II-B & C. Equivalent ratios

💡 Equivalent ratios: multiply or divide BOTH parts by the same number.

28. For every 3 ibises the birdwatcher sees 5 sunbirds. What is the ratio of ibises to birds?

A 3:5 B 5:3
C 3:8 ✓ D 5:8

Why: Birds = 3 + 5 = 8 → ibises : birds = 3 : 8.

29. Ibises to sunbirds is 3:5. Complete the equivalent ratio 6 : ___

A 8 **B 10 ✓**
C 9 D 15

Why: 3 × 2 = 6, so 5 × 2 = 10.

30. Ibises to sunbirds is 3:5. Complete the equivalent ratio 9 : ___

A 12 B 10
C 15 ✓ D 18

Why: 3 × 3 = 9, so 5 × 3 = 15.

31. If the birdwatcher sees 25 sunbirds, how many ibises does she see?

A 15 ✓ B 10
C 20 D 25

Why: 25 ÷ 5 = 5, then 5 × 3 = 15 ibises.

32. A mix of concrete uses 5 parts sand and 3 parts cement. Write the ratio of sand to concrete.

A 5:3 B 3:5
C 5:8 ✓ D 8:5

Why: Concrete = 5 + 3 = 8 parts → sand : concrete = 5 : 8.

33. Milan has 20 bags of sand (ratio sand : cement = 5 : 3). How many bags of cement will he use?

A 20:8 **B 20:12 ✓**
C 20:15 D 20:3

Why: 20 ÷ 5 = 4, then 4 × 3 = 12 bags of cement → 20 : 12.

II-D. Use the graph (races won vs. lost)

The graph shows points at (5, 1), (10, 2), (15, 3) and (20, 4): for every 5 races won, the runner lost 1 race.

💡 Read one point from the graph (5 wins, 1 loss) and multiply both numbers to find equivalent ratios.

34. Complete the ratio of wins to losses: ___ : 1

A 4 **B 5 ✓**
C 10 D 20

Why: The first point is (5 won, 1 lost) → 5 : 1.

35. Complete the equivalent ratio: 5 : ___

A 1 ✓ B 2
C 5 D 10

36. Complete the equivalent ratio: ___ : 2

A 5 **B 10 ✓**
C 15 D 20

Why: 5 × 2 = 10 wins for 2 losses.

37. Complete the equivalent ratio: ___ : 3

A 10 **B 15 ✓**
C 20 D 9

Why: 5 × 3 = 15 wins for 3 losses.

38. Complete the equivalent ratio: 20 : ___

A 2 B 3
C **4** ✓ D 5

Why: 20 ÷ 5 = 4 losses.

39. If the runner won 10 races, how many did he lose?

A 1 B **2** ✓
C 5 D 10

Why: 10 ÷ 5 = 2 losses.

II-E. Decimal sums and differences

💡 Line up the decimal points, add zeros to fill empty places, then add or subtract.

40. 9.4 + 83.2 =

A **92.6** ✓ B 92.4
C 93.6 D 91.6

41. 65.329 − 42.49 =

A **22.839** ✓ B 23.839
C 22.939 D 22.84

Why: 65.329 − 42.490 = 22.839.

42. 17.42 + 3.067 =

A **20.487** ✓ B 20.387
C 21.487 D 20.427

Why: 17.420 + 3.067 = 20.487.

43. 192.022 − 148.347 =

A **43.675** ✓ B 44.675
C 43.775 D 43.325

44. 4.056 + 87.06 =

A **91.116** ✓ B 91.016
C 92.116 D 91.126

Why: 4.056 + 87.060 = 91.116.

45. 783.29 + 89.0054 =

A **872.2954** ✓ B 872.3954
C 871.2954 D 872.2854

Why: 783.2900 + 89.0054 = 872.2954.

46. 357.04 − 254.769 =

- **A 102.271 ✓**
- B 102.371
- C 103.271
- D 102.281

Why: 357.040 − 254.769 = 102.271.

47. 32.056 − 25.8407 =

- **A 6.2153 ✓**
- B 6.3153
- C 7.2153
- D 6.2163

Why: 32.0560 − 25.8407 = 6.2153.

II-F. Decimal products

💡 Multiply as whole numbers, then count the decimal places in both factors and put the point back.

48. 22 × 3.7 =

- **A 81.4 ✓**
- B 80.4
- C 8.14
- D 814

Why: 22 × 37 = 814; one decimal place → 81.4.

49. 5 × 2.562 =

- **A 12.810 ✓**
- B 12.710
- C 13.810
- D 1.281

Why: 5 × 2562 = 12810; three decimal places → 12.810.

50. 12.9 × 2.35 =

- **A 30.315 ✓**
- B 30.215
- C 31.315
- D 3.0315

Why: 129 × 235 = 30315; three decimal places → 30.315.

51. 6.88 × 7.32 =

- **A 50.3616 ✓**
- B 50.2616
- C 49.3616
- D 503.616

Why: 688 × 732 = 503616; four decimal places → 50.3616.

II-G. Quotients and remainders

💡 Long division: divide, multiply, subtract, bring down. The remainder must be smaller than the divisor.

52. 472 ÷ 8 =

- **A 59 ✓**
- B 58
- C 59 R4
- D 69

Why: 8 × 59 = 472 exactly.

53. 2,175 ÷ 29 =

A 75 ✓ **B** 74

C 75 R5 **D** 85

Why: 29 × 75 = 2,175 exactly.

54. 20,098 ÷ 54 =

A 372 R10 ✓ **B** 372

C 371 R10 **D** 372 R18

Why: 54 × 372 = 20,088; 20,098 − 20,088 = 10 → 372 R10.

55. 4,813 ÷ 332 =

A 14 R165 ✓ **B** 14

C 15 R165 **D** 13 R165

Why: 332 × 14 = 4,648; 4,813 − 4,648 = 165 → 14 R165.

II-H & I. Multiply and divide fractions

💡 Change mixed numbers to improper fractions. To divide, multiply by the reciprocal (flip the second fraction). Simplify at the end.

56. 3 2/3 × 6 × 7/12 = (simplest form)

A 12 5/6 ✓ **B** 12 1/6

C 11 5/6 **D** 13 5/6

Why: 11/3 × 6/1 × 7/12 = 77/6 = 12 5/6.

57. 7 × 3 3/7 × 5/14 = (simplest form)

A 8 4/7 ✓ **B** 8 3/7

C 7 4/7 **D** 8 4/14

Why: 7/1 × 24/7 × 5/14 = 120/14 = 60/7 = 8 4/7.

58. 1 1/5 × 1 1/6 × 1 2/7 = (simplest form)

A 1 4/5 ✓ **B** 1 3/5

C 2 4/5 **D** 1 4/7

Why: 6/5 × 7/6 × 9/7 = 9/5 = 1 4/5.

59. 8 2/5 ÷ 3 = (simplest form)

A 2 4/5 ✓ **B** 2 2/5

C 25 1/5 **D** 3 4/5

Why: 42/5 × 1/3 = 14/5 = 2 4/5.

60. **5/8 ÷ 15/22 = (simplest form)**

A **11/12** ✓ | **B** 12/11

C 75/176 | **D** 5/12

Why: 5/8 × 22/15 = 110/120 = 11/12.

61. **4/7 ÷ 14 = (simplest form)**

A **2/49** ✓ | **B** 4/98

C 8 | **D** 1/49

Why: 4/7 × 1/14 = 4/98 = 2/49.

II-J. Volume of rectangular prisms

💡 Volume = length × width × height. Change mixed numbers to improper fractions first.

62. **A rectangular prism measures 2 1/2 in × 5 in × 1 1/4 in. Its volume is ...**

A **15 5/8 in³** ✓ | **B** 15 1/8 in³

C 8 3/4 in³ | **D** 16 5/8 in³

Why: 5/2 × 5 × 5/4 = 125/8 = 15 5/8 in³.

63. **A rectangular prism measures 7 1/2 m × 4 1/2 m × 6 3/5 m. Its volume is ...**

A **222 3/4 m³** ✓ | **B** 222 1/4 m³

C 221 3/4 m³ | **D** 18 3/5 m³

Why: 15/2 × 9/2 × 33/5 = 4455/20 = 222 3/4 m³.

II-K. Solve

💡 'Of' means multiply (3/5 of 200 = 3/5 × 200). Work out each item's cost, then add.

64. **A bakery made 200 packages of dinner rolls. 3/5 of them were sold to restaurants. How many packages were sold to restaurants?**

A **120** ✓ | **B** 80

C 60 | **D** 150

Why: 3/5 × 200 = 120.

65. **Of the 200 packages, how many were sold at the bakery?**

A 120 | **B** **80** ✓

C 100 | **D** 40

Why: 200 − 120 = 80.

66. **Wes bought 2 notebooks for $1.79 each and 3 pens for $0.69 each. How much did he spend in all?**

A $5.65 ✓

B $2.48

C $5.56

D $4.96

Why: 2 × 1.79 = 3.58; 3 × 0.69 = 2.07; 3.58 + 2.07 = $5.65.